You check your phone and see a mobile money transfer of `$25`, a weather app showing `32°C`, and a message from school about your attendance. Each of those is a piece of data, a small fact about something that happened. Before you analyze data, build dashboards, or choose a career in data, you need a clear answer to a simple question: **what is data?**

CHAPTER 1 — WHAT IS DATA?

What Is Data?

Lesson 1 of 5

BY THE END

You can explain what data is, and tell data, information, and insight apart.

Data: raw facts about the world

Data is a collection of facts, measurements, or observations about the world. On its own, data usually does not tell a full story. It is the raw material we use to understand what is happening.

Examples of data

- `$25` — a mobile money transfer amount
- `$2.50 per kg` — the market price of rice
- `present` — a student’s attendance record
- `$49.99` — a product price
- `32°C` — a temperature reading

Each example is a single fact. Collect enough of them and patterns start to appear.

Data vs information vs insight

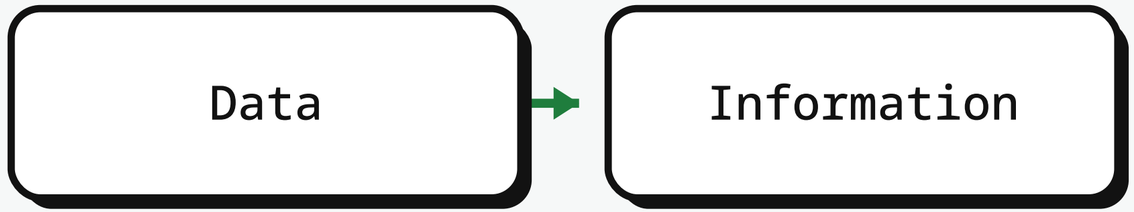
These three terms are related, but they are not the same.

TERM	MEANING	EXAMPLE
`Data`	Raw facts	Sales were `$1,200`, `$980` and `$1,450` on three days
`Information`	Data organized to answer a basic question	Sales increased from Monday to Wednesday
`Insight`	A useful conclusion that can guide action	Promotions on Wednesday may be driving higher sales

PLATE I — FROM FACTS TO A DECISION

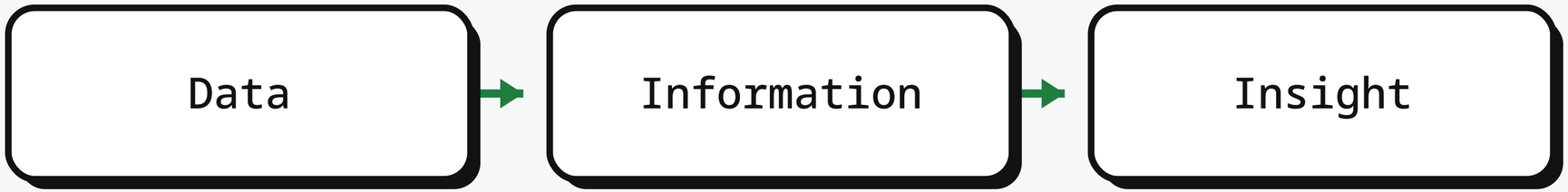
Raw facts

PLATE I — FROM FACTS TO A DECISION



Data organized to answer a basic question

PLATE I — FROM FACTS TO A DECISION



Each step throws information away and adds meaning. You cannot skip one: an insight with no data underneath it is just an opinion.

A useful conclusion that can guide action

Think of it like cooking

DATA	INFORMATION	INSIGHT
The ingredients, before anyone has decided what to make.	A prepared meal you can actually understand.	Knowing what to cook next time, based on what people liked.

Where you see data every day

Data is not only found in spreadsheets or databases. You create and use data constantly:

- Mobile money transactions and bank transfers
- School grades and attendance records
- Market prices at a local shop
- Likes and comments on social media
- Steps tracked by a fitness app
- Weather forecasts

Businesses, schools, hospitals, and governments all rely on data to make decisions.

KEY TERMS

TERM	DEFINITION
Data	Raw facts, measurements, or observations
Information	Data that has been organized to convey meaning
Insight	A meaningful conclusion, supported by data, that helps guide a decision